

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (EE)

May 2012

EE306 Electrical Machines III

Date: 07.05.2012, Monday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 Draw and explain the capability curve of synchronous generator. Also explain the power limits. [07]
- Q - 2 (a) A 3 phase, 4 pole, 50 Hz, star connected alternator has 60 slots with 2 conductors per slot and having armature winding of the double layer type. Coils are short pitched, that is, if one coil side lies in slot number 1, the other coil side lies in slot number 13. Find the useful flux per pole required to induce a line voltage of 6.6kV. [07]
- (b) Define armature reaction of an alternator Explain the effect of armature reaction on the terminal voltage of an alternator at (i) unity power factor load (ii) zero leading power factor load (iii) zero lagging power factor load. Draw the relevant phasor diagrams. [07]

OR

- Q - 2 (a) Discuss the condition of Synchronization of a three phase alternator to infinite bus bar. [07]
- (b) A 3 MVA, 6 pole alternator runs at 1000 rpm on 3.3 kV busbars. The synchronous reactance is 25 %. Calculate the synchronizing power and torque per mechanical degree of displacement when the alternator is supplying full load at 0.8 power factor lag. [07]
- Q - 3 (a) Explain the Potier method of finding the voltage regulation of an alternator. [07]
- (b) Discuss the slip test for measurement of X_d and X_q in a salient pole synchronous machine. [07]

OR

- Q - 3 (a) Draw and explain the phasor diagram of a salient pole synchronous generator supplying full load lagging current. Derive the power output equation
$$P = (3V E_f / X_d) \sin \delta + 3(V^2 / 2) [1 / X_d - 1 / X_q] \sin 2\delta.$$
 [07]
- (b) List the various losses that occur in an alternator. Draw the power flow diagram. Discuss a suitable method of determining losses and efficiency of alternator. [07]

SECTION – II

Q - 4 Draw and explain the families of V curves and inverted V curves as applied to synchronous motor. [07]

Q - 5 (a) A 75 kW, 400 V, 3 phase, 4 pole, star connected synchronous motor has a resistance and synchronous reactance per phase of 0.04 ohm and 0.4 ohm respectively. Compute for full load 0.8 power factor lead the open circuit emf per phase and gross mechanical power developed. Assume the efficiency of 92.5 %. [06]

(b) Explain the function of damper winding in synchronous machines. [04]

(c) Explain the principle of operation of 3 phase synchronous motor and means to start it. [04]

OR

Q - 5 (a) Explain the synchronizing current, synchronizing power and synchronizing torque when two alternators are connected in parallel and derive its expression. [07]

(b) A 3 phase synchronous motor absorbing 60 kW is connected in parallel with a factory load of 240 kW having a lagging power factor of 0.8. If the combined load has a power factor of 0.9 lagging, what is the value of leading kVAR supplied by the motor and at what power factor is it working? [07]

Q - 6 (a) Write a note on Synchronous machines transient reactance and time constants [07]

(b) Explain factors affecting the transient stability and methods for improving the transient stability. [07]

OR

Q - 6 (a) Explain Permanent Magnet Brushless DC (PMBLDC) and Permanent Magnet Synchronous machine. [07]

(b) Explain construction, working and application of Switched Reluctance Motor (SRM). [07]
